Populations and phenotypes

GB Goodman–Buckler panel

280 inbred lines

Grain free amino acid GWAS (71 traits)

Free pool dominated by Asn and Pro

SN Maize landraces

3,107 georeferenced accessions

Environmental GWAS of soil N

12 SNPs pass specificity tests

IC Indian Chief, C0 vs C14

Reciprocal recurrent selection

FST scan, top-5% windows

Free AA and Pro up, Asp and Glu down

J Jarvis, C0 vs C14

Reciprocal recurrent selection

FST scan, top-5% windows

Asp fraction up (nominal only)

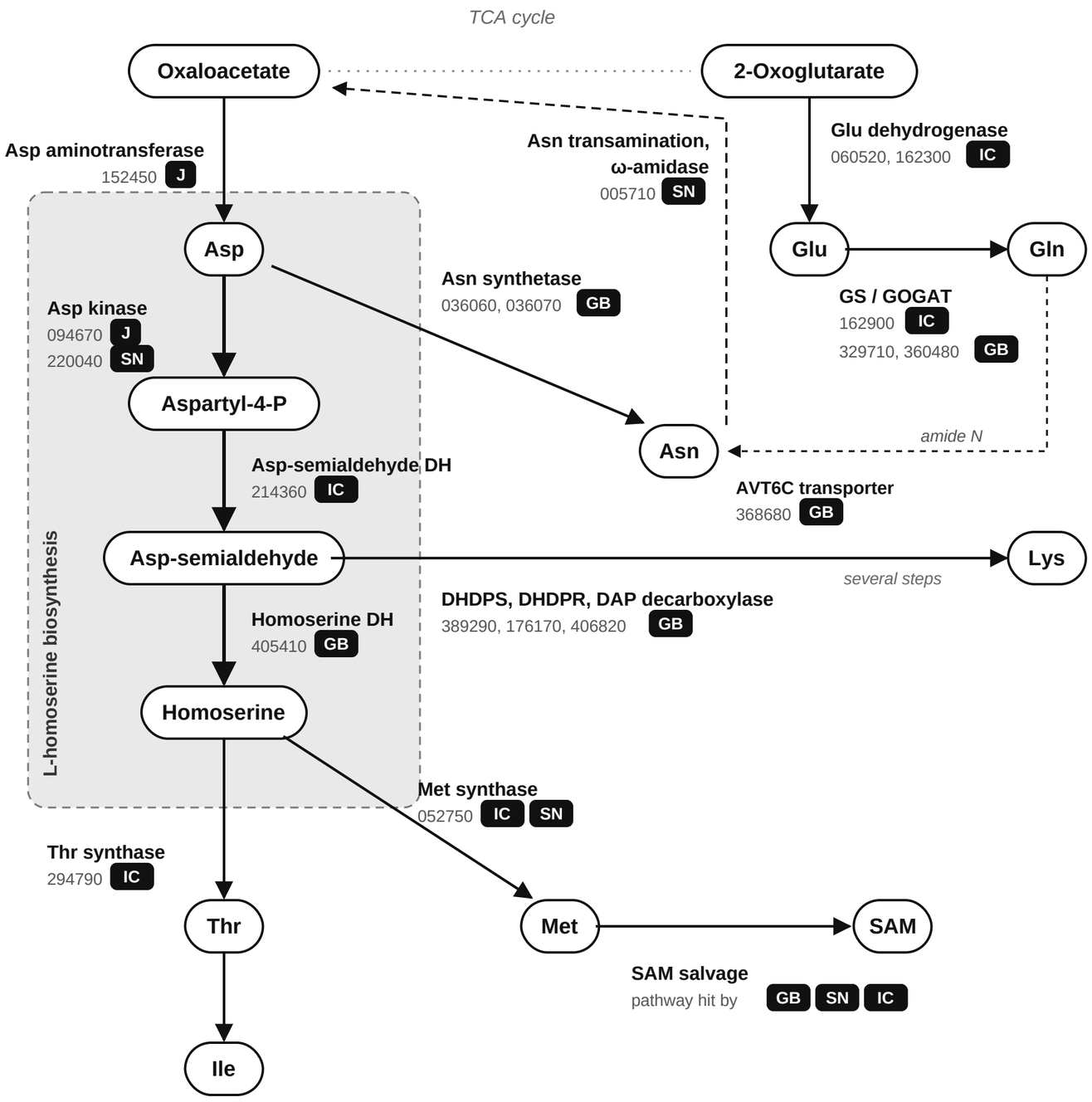
QTL B64 × CML14 $F_{2:3}$

Biparental cross

QTL mapping of free Asp and Pro

Shared Asp/Pro QTL on chr 3

Candidate genes on aspartate and glutamate metabolism



Other candidate genes

(outside the aspartate pathway)

Protein turnover

- GB Peptidase A1 (190800), 8 traits
- IC SAG39 cysteine protease (214780)
- IC ATG18a-like autophagy (352300)
- J NPL4-like (146560)

ER membrane proteins

- GB EMC subunit 4 (394360)
- GB Rhomboid-like 19 (394380), 9 traits

Transport

- GB NPF4.6, UmamiTs, AVT3A
- J LHT-like, ProT-like
- IC AMT2-like (214890)

Genes are positional candidates: within 25 kb of a significant GWAS SNP, or in a top-5% FST window. Numbers are Zm00001eb gene IDs (B73 RefGen_v5).

Shaded box: L-homoserine biosynthesis, the only CornCyc pathway with a candidate gene in all four analyses (permutation P = 0.0012; not significant after correction for 468 pathways).

Thick arrows, steps of L-homoserine biosynthesis. Dashed arrows, amide-N donation from Gln to Asn synthesis, and Asn transamination with ω-amidase returning carbon to oxaloacetate.